

How to Identify Fumbling to Keep a Network Secure

Repeated systematic failed attempts by a host to access resources like a URL, an IP address or an email address is known as fumbling. Erroneous attempts to access resources by legitimate users must not be confused with fumbling. Let's look at how we can design an effective system for identifying network fumbling, to help keep our networks secure.



Network security implementation mainly depends on exploratory data analysis (EDA) and visualisation. EDA provides a mechanism to examine a data set without preconceived assumptions about the data and its behaviour. The behaviour of the Internet and the attackers is dynamic and EDA is a continuous process to help identify all the phenomena that are cause for an alarm, and to help detect anomalies in access to resources.

Fumbling is a general term for repeated systematic failed attempts by a host to access resources. For example, legitimate users of a service should have a valid email ID or user identification. So if there are numerous attempts by a user from a different location to target the users of this service with different email identifications, then there is a chance that this is an attack from that location. From the data analysis point of view, we say a fumbling condition has happened.

This indicates that the user does not have access to that system and is exploring different possibilities to break the

security of the target. It is the task of the security personnel to identify the pattern of the attack and the mistakes committed to differentiate them from innocent errors. Let's now discuss a few examples to identify a fumbling condition.

In a nutshell, fumbling is a type of Internet attack, which is characterised by failing to connect to one location with a systematic attack from one or more locations. After a brief discussion of this type of network intrusion, let's consider a problem of network data analysis using R, which is a good choice as it provides powerful statistical data analysis tools together with a graphical visualisation opportunity for a better understanding of the data.

Fumbling of the network and services

In case of TCP fumbling, a host fails to reach a target port of a host, whereas in the case of HTTP fumbling, hackers fail to access a target URL. All fumbling is not a network attack, but most of the suspicious attacks appear as fumbling.